

Metrics Definitions

Enterprise Analytics Knowledge Base — TPC-H Data Platform

This document defines the key business metrics used across the enterprise analytics platform. It is the authoritative reference for how each KPI is calculated, which database tables and columns are involved, and how results should be interpreted. All metrics are derived from the TPC-H dataset hosted on Databricks (catalog: **samples**, schema: **tpch**).

1. Revenue Metrics

1.1 Total Revenue

Definition: The sum of all extended prices charged to customers, after applying discounts, for all shipped line items in a given period.

Formula: $\text{SUM}(l_extendedprice * (1 - l_discount))$

Tables used: samples.tpch.lineitem, samples.tpch.orders

Key columns:

- l_extendedprice: the base price of the line item
- l_discount: fractional discount applied (0.00 to 0.10)
- l_shipdate: used to filter by period
- o_orderdate: alternative date filter at order level

Example query: What was total revenue by nation last year?

1.2 Average Order Value (AOV)

Definition: The average monetary value of a single customer order, calculated by dividing total revenue by the number of distinct orders.

Formula: $\text{SUM}(l_extendedprice * (1 - l_discount)) / \text{COUNT}(\text{DISTINCT } o_orderkey)$

Tables used: samples.tpch.lineitem, samples.tpch.orders

Interpretation: A rising AOV indicates customers are purchasing higher-value items or placing larger orders. It is a primary indicator of sales quality.

Example query: What is the Average Order Value per market segment?

1.3 Revenue by Region / Nation

Definition: Total revenue broken down by geographic region or nation, used for regional performance comparison and territory management.

Tables used: lineitem, orders, customer, nation, region

Join path: lineitem → orders (o_orderkey) → customer (o_custkey) → nation (c_nationkey) → region (n_regionkey)

2. Order Metrics

2.1 Order Count

Definition: The total number of distinct orders placed in a given period.

Formula: COUNT(DISTINCT o_orderkey)

Tables used: samples.tpch.orders

2.2 Order Fulfillment Rate

Definition: The percentage of orders where all line items have been shipped, indicating operational efficiency of the supply chain.

Formula: COUNT(orders where all items shipped) / COUNT(total orders) * 100

Tables used: samples.tpch.orders, samples.tpch.lineitem

Order status values: O = Open, F = Fulfilled, P = Partially shipped

2.3 Average Days to Ship

Definition: The average number of days between an order being placed and the line items being shipped. A key supply chain performance metric.

Formula: AVG(DATEDIFF(l_shipdate, o_orderdate))

Tables used: samples.tpch.lineitem, samples.tpch.orders

3. Supplier and Cost Metrics

3.1 Supply Cost

Definition: The total cost paid to suppliers for parts, calculated from the partsupp table which records the cost per supplier per part.

Formula: SUM(ps_supplycost * ps_availqty)

Tables used: samples.tpch.partsupp, samples.tpch.supplier

3.2 Supplier Performance

Definition: Ranking of suppliers by volume of revenue they contributed through fulfilled line items. Used for vendor management decisions.

Formula: SUM(l_extendedprice * (1 - l_discount)) grouped by s_name

Tables used: samples.tpch.lineitem, samples.tpch.supplier

4. Quick Reference Table

Metric	Formula	Primary Table	Role Access
Total Revenue	SUM(extendedprice*(1-discount))	lineitem	Finance, Executive
Avg Order Value	Total Revenue / Order Count	lineitem + orders	Analyst, Finance, Executive
Order Count	COUNT(DISTINCT orderkey)	orders	Analyst, Executive
Fulfillment Rate	Fulfilled / Total * 100	orders	Analyst, Executive
Avg Days to Ship	AVG(shipdate - orderdate)	lineitem	Finance, Executive
Supply Cost	SUM(supplycost * availqty)	partsupp	Finance, Executive
Revenue by Region	Total Revenue by n_name/r_name	multi-table join	Executive

5. Date Filtering Convention

The TPC-H dataset covers order dates from **1992 to 1998**. When users ask about 'last year', 'this year', or 'recent' data, interpret relative to the dataset range and use 1997–1998 as the most recent period. Always filter using **o_orderdate** for order-level queries and **l_shipdate** for shipment-level queries.

Example: 'Last year' = WHERE YEAR(o_orderdate) = 1997